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**Q1. String Character Frequency Counter**

CODE

```
text = input().lower().replace(" ", "")
counts = {c: text.count(c) for c in text}
for char in sorted(counts):
    print(f"{char} = {counts[char]}")
```

INPUT

Hello World

OUTPUT

```
d = 1
e = 1
h = 1
l = 3
o = 2
r = 1
w = 1
```

**Q2. Vowel and Consonant Classifier with Statistics**

CODE

```
sentence = input("Enter a sentence: ").lower()

vowels = "aeiouAEIOU"
vowel_count = 0
consonant_count = 0
vowel_freq = {}
consonant_freq = {}

for ch in sentence:
    if ch.isalpha():
        if ch in vowels:
            vowel_count += 1
            vowel_freq[ch] = vowel_freq.get(ch, 0) + 1
        else:
            consonant_count += 1
            consonant_freq[ch] = consonant_freq.get(ch, 0) + 1

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)

if vowel_freq:
    mv = max(vowel_freq, key=vowel_freq.get)
    print("Most frequent vowel:", mv, "(", vowel_freq[mv], ")", sep="")
else:
    print("Most frequent vowel: None")

if consonant_freq:
    lc = min(consonant_freq, key=consonant_freq.get)
    print("Least frequent consonant:", lc, "(", consonant_freq[lc], ")", sep="")
else:
    print("Least frequent consonant: None")

if consonant_count > 0:
    print("Vowel-to-Consonant Ratio:", round(vowel_count / consonant_count, 2))
else:
    print("Vowel-to-Consonant Ratio: Undefined")
```

INPUT

Python Programming

OUTPUT

```
Enter a sentence: Vowels: 4
Consonants: 13
Most frequent vowel:o(2)
Least frequent consonant:y(1)
Vowel-to-Consonant Ratio: 0.31
```

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**Q3. String Compression and Run-Length Encoding****CODE**

```
def compress(s):
    if not s:
        return ""

    comp = ""
    count = 1

    for i in range(1, len(s)):
        if s[i] == s[i - 1]:
            count += 1
        else:
            comp += s[i - 1]
            if count > 1:
                comp += str(count)
            count = 1

    comp += s[-1]
    if count > 1:
        comp += str(count)

    if len(comp) < len(s):
        return comp
    else:
        return s

def decompress(s):
    result = ""
    i = 0

    while i < len(s):
        ch = s[i]
        i += 1
        num = ""

        while i < len(s) and s[i].isdigit():
            num += s[i]
            i += 1

        if num == "":
            result += ch
        else:
            result += ch * int(num)

    return result

text = input("Enter string: ")

compressed = compress(text)

if compressed == text:
    print("Compression not beneficial.")
    print("Original:", text)
else:
    print("Compressed:", compressed)
    print("Decompressed:", decompress(compressed))
```

**INPUT** **OUTPUT**

```
Enter string: Compression not beneficial.
Original: abc
```

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**Q4. Advanced Text Statistics Analyser**

**CODE**

```
import re

text = input("Enter paragraph: ")

paragraphs = 1 if text.strip() else 0

sentences = len(re.findall(r'[.!?]', text))
if sentences == 0 and text.strip():
    sentences = 1

words = re.findall(r'\b[a-zA-Z]+\b', text.lower())
total_words = len(words)

avg_word = sum(len(w) for w in words) / total_words if total_words else 0
avg_sentence = total_words / sentences if sentences else 0

stop = {"the", "is", "a", "an", "in", "of", "and", "to", "for"}

freq = {}
for w in words:
    if w not in stop:
        freq[w] = freq.get(w, 0) + 1

top = sorted(freq.items(), key=lambda x: (-x[1], x[0]))[:5]

vowels = "aeiou"
syllables = 0

for word in words:
    count = 0
    prev = False
    for ch in word:
        if ch in vowels:
            if not prev:
                count += 1
            prev = True
        else:
            prev = False
    if count == 0:
        count = 1
    syllables += count

if total_words and sentences:
    flesch = 206.835 - 1.015 * (total_words / sentences) - 84.6 * (syllables / total_words)
else:
    flesch = 0

print("Words:", total_words)
print("Sentences:", sentences)
print("Paragraphs:", paragraphs)
print("Avg Word Length:", round(avg_word, 2))
print("Avg Sentence Length:", round(avg_sentence, 2))

print("Top 5 Words:")
for w, c in top:
    print(w, "(", c, ")", sep=" ")

print("Flesch Reading Ease:", round(flesch, 2))
```

**INPUT**

Python is a powerful programming language

**OUTPUT**

Enter paragraph: Words: 6  
 Sentences: 1  
 Paragraphs: 1  
 Avg Word Length: 6.0  
 Avg Sentence Length: 6.0  
 Top 5 Words:  
 language(1)  
 powerful(1)